

Dot product and Vector angles

Meenu S

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Dot Product (Scalar Product)

Let

$\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$ be two vectors in $\mathbb{R}^n$.

Definition

The **dot product** (or scalar product) of $\mathbf{a}$ and $\mathbf{b}$ is defined as

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + \dots + a_nb_n = \sum_{i=1}^n a_ib_i.$$

Geometric Interpretation

If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta,$$

where $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$ are the magnitudes (norms) of the vectors.

Properties

1. Commutative: $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
2. Distributive over addition: $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
3. Scalar multiplication: $(k\mathbf{a}) \cdot \mathbf{b} = k(\mathbf{a} \cdot \mathbf{b})$ for any scalar k
4. Positive-definite: $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \geq 0$, and equality holds if and only if $\mathbf{a} = \mathbf{0}$

Applications

- Finding the angle between two vectors: $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$
- Checking orthogonality: $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$
- Projection of $\mathbf{a}$ on $\mathbf{b}$: $\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \mathbf{b}$

1) Given $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -3)$, find:

1. $\mathbf{u} \cdot \mathbf{v}$

$$\mathbf{u} \cdot \mathbf{v} = 3 \cdot 2 + 4 \cdot (-3) = 6 - 12 = -6$$

2. $\mathbf{u} \cdot \mathbf{u}$

$$\mathbf{u} \cdot \mathbf{u} = 3^2 + 4^2 = 9 + 16 = 25$$

3. $\|\mathbf{u}\|^2$

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 25$$

4. $(\mathbf{u} \cdot \mathbf{v})\mathbf{v}$

$$(\mathbf{u} \cdot \mathbf{v})\mathbf{v} = (-6)(2, -3) = (-12, 18)$$

5. $\mathbf{u} \cdot (5\mathbf{v})$

$$\mathbf{u} \cdot (5\mathbf{v}) = 5(\mathbf{u} \cdot \mathbf{v}) = 5(-6) = -30$$

2) Given $\mathbf{u} = (-1, 1, -2)$ and $\mathbf{v} = (1, -3, -2)$, find:

1. $\mathbf{u} \cdot \mathbf{v}$

$$\mathbf{u} \cdot \mathbf{v} = (-1)(1) + (1)(-3) + (-2)(-2) = -1 - 3 + 4 = 0$$

2. $\mathbf{u} \cdot \mathbf{u}$

$$\mathbf{u} \cdot \mathbf{u} = (-1)^2 + 1^2 + (-2)^2 = 1 + 1 + 4 = 6$$

3. $\|\mathbf{u}\|^2$

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = 6$$

4. $(\mathbf{u} \cdot \mathbf{v})\mathbf{v}$

$$(\mathbf{u} \cdot \mathbf{v})\mathbf{v} = 0 \cdot (1, -3, -2) = (0, 0, 0)$$

5. $\mathbf{u} \cdot (5\mathbf{v})$

$$\mathbf{u} \cdot (5\mathbf{v}) = 5(\mathbf{u} \cdot \mathbf{v}) = 5 \cdot 0 = 0$$

3) Given $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (-2, 4)$, find the angle θ between them.

$$\text{Recall: } \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Step 1: Compute the dot product

$$\mathbf{u} \cdot \mathbf{v} = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2$$

Step 2: Compute the magnitudes

$$\|\mathbf{u}\| = \sqrt{3^2 + 1^2} = \sqrt{9 + 1} = \sqrt{10}, \quad \|\mathbf{v}\| = \sqrt{(-2)^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20}$$

Step 3: Compute $\cos \theta$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{-2}{\sqrt{10} \cdot \sqrt{20}} = \frac{-2}{\sqrt{200}} = \frac{-2}{10\sqrt{2}} = -\frac{1}{5\sqrt{2}}$$

Step 4: Find θ

$$\theta = \cos^{-1} \left(-\frac{1}{5\sqrt{2}} \right)$$

4) Given $\mathbf{u} = (1, 1, 1)$ and $\mathbf{v} = (2, 1, -1)$, find the angle θ between them.

Step 1: Compute the dot product

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 2 + 1 \cdot 1 + 1 \cdot (-1) = 2 + 1 - 1 = 2$$

Step 2: Compute the magnitudes

$$\|\mathbf{u}\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}, \quad \|\mathbf{v}\| = \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{4 + 1 + 1} = \sqrt{6}$$

Step 3: Compute $\cos \theta$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{2}{\sqrt{3} \cdot \sqrt{6}} = \frac{2}{\sqrt{18}} = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}$$

Step 4: Find θ

$$\theta = \cos^{-1} \left(\frac{\sqrt{2}}{3} \right)$$

5) Vectors orthogonal to $\mathbf{u} = (0, 5)$

Let $\mathbf{v} = (x, y)$ be orthogonal to $\mathbf{u}$.

$$\mathbf{u} \cdot \mathbf{v} = 0 \implies 0 \cdot x + 5 \cdot y = 0 \implies y = 0$$

Hence all vectors orthogonal to $\mathbf{u}$ are of the form $\mathbf{v} = (x, 0)$, $x \in \mathbb{R}$.

6) Vectors orthogonal to $\mathbf{u} = (4, -1, 0)$

Let $\mathbf{v} = (x, y, z)$ be orthogonal to $\mathbf{u}$.

$$\mathbf{u} \cdot \mathbf{v} = 0 \implies 4x + (-1)y + 0 \cdot z = 0 \implies 4x - y = 0 \implies y = 4x$$

Hence all vectors orthogonal to $\mathbf{u}$ are of the form $\mathbf{v} = (x, 4x, z)$, $x, z \in \mathbb{R}$.

7) Given $\mathbf{u} = (2, 18)$, $\mathbf{v} = (\frac{3}{2}, -\frac{1}{6})$, determine whether they are orthogonal, parallel, or neither.

Step 1: Check orthogonality

$$\mathbf{u} \cdot \mathbf{v} = 2 \cdot \frac{3}{2} + 18 \cdot \left(-\frac{1}{6}\right) = 3 - 3 = 0$$

Since $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors are orthogonal.

8) Given $\mathbf{u} = (0, 1, 0)$, $\mathbf{v} = (1, -2, 0)$, determine whether they are orthogonal, parallel, or neither.

Step 1: Check orthogonality

$$\mathbf{u} \cdot \mathbf{v} = 0 \cdot 1 + 1 \cdot (-2) + 0 \cdot 0 = -2$$

Since $\mathbf{u} \cdot \mathbf{v} \neq 0$, the vectors are not orthogonal.

Step 2: Check parallelism

$$\mathbf{u} = (0, 1, 0), \quad \mathbf{v} = (1, -2, 0)$$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{0 \cdot 1 + 1 \cdot (-2) + 0 \cdot 0}{\sqrt{0^2 + 1^2 + 0^2} \sqrt{1^2 + (-2)^2 + 0^2}} = \frac{-2}{1 \cdot \sqrt{5}} = -\frac{2}{\sqrt{5}}$$

Since $|\cos \theta| = \frac{2}{\sqrt{5}} \neq 1$, the vectors are not parallel.

The vectors are neither orthogonal nor parallel.

9) Explain what is known about θ , the angle between $\mathbf{u}$ and $\mathbf{v}$, if:

1. $\mathbf{u} \cdot \mathbf{v} = 0$
2. $\mathbf{u} \cdot \mathbf{v} > 0$
3. $\mathbf{u} \cdot \mathbf{v} < 0$

Solution:

The angle θ between two vectors $\mathbf{u}$ and $\mathbf{v}$ is related to the dot product by

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

1. **If $\mathbf{u} \cdot \mathbf{v} = 0$:**

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta = 0$$

Since $\|\mathbf{u}\| \neq 0$ and $\|\mathbf{v}\| \neq 0$, it must be that

$$\cos \theta = 0 \implies \theta = 90^\circ$$

Hence, the vectors are orthogonal (perpendicular to each other).

2. **If $\mathbf{u} \cdot \mathbf{v} > 0$:**

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta > 0$$

Therefore,

$$\cos \theta > 0 \implies 0^\circ < \theta < 90^\circ$$

Hence, the angle between the vectors is acute.

3. **If $\mathbf{u} \cdot \mathbf{v} < 0$:**

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta < 0$$

Therefore,

$$\cos \theta < 0 \implies 90^\circ < \theta < 180^\circ$$

Hence, the angle between the vectors is obtuse.

Inner Product

Inner Product (Real Vector Spaces)

Let V be a vector space over $\mathbb{R}$. An *inner product* on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

satisfying the following axioms for all $u, v, w \in V$ and $\alpha \in \mathbb{R}$:

1. **Symmetry:**

$$\langle u, v \rangle = \langle v, u \rangle$$

2. **Additivity (Linearity in the first argument):**

$$\langle u + w, v \rangle = \langle u, v \rangle + \langle w, v \rangle$$

3. **Homogeneity (Scalar multiplication in the first argument):**

$$\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$$

4. **Positive Definiteness:**

$$\langle v, v \rangle \geq 0 \quad \text{and} \quad \langle v, v \rangle = 0 \iff v = 0$$

- The norm (or length) of a vector v is defined using the inner product as

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

Properties:

- $\|v\| \geq 0$ and $\|v\| = 0 \iff v = 0$ (positive definiteness)
- $\|\alpha v\| = |\alpha| \|v\|$ for any scalar α
- The distance between two vectors u and v is defined as

$$d(u, v) = \|u - v\| = \sqrt{\langle u - v, u - v \rangle}.$$

This satisfies all the usual properties of a distance function.

- If $u, v \neq 0$, the angle θ between u and v is defined by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}, \quad 0 \leq \theta \leq \pi.$$

Remarks:

- $\langle u, v \rangle > 0 \implies$ acute angle
- $\langle u, v \rangle = 0 \implies$ right angle
- $\langle u, v \rangle < 0 \implies$ obtuse angle

Note: Complex Vector Spaces

If V is a vector space over $\mathbb{C}$, the inner product is usually defined to satisfy:

1. Conjugate Symmetry:

$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

2. Linearity and Homogeneity in the first argument:

$$\langle \alpha u + w, v \rangle = \alpha \langle u, v \rangle + \langle w, v \rangle$$

3. Positive Definiteness:

$$\langle v, v \rangle \geq 0, \quad \langle v, v \rangle = 0 \iff v = 0$$

Remark: In complex spaces, the inner product is only linear in the first argument; in the second argument it is conjugate-linear.

Let V be a real inner product space (for complex spaces, see notes below), and let $u, v \in V$. If V is a complex inner product space:

- The norm and distance formulas are the same:

$$\|v\| = \sqrt{\langle v, v \rangle}, \quad d(u, v) = \|u - v\|$$

- The inner product $\langle u, v \rangle$ may be complex. Hence, the cosine of the angle is often defined using the real part:

$$\cos \theta = \frac{\operatorname{Re}(\langle u, v \rangle)}{\|u\| \|v\|}.$$

- This ensures that θ is a real number and lies between 0 and π .

Projection of a Vector

Let $u, v \in V$ with $v \neq 0$.

Real Inner Product Spaces: The projection of u onto v is defined as

$$\operatorname{proj}_v(u) = \frac{\langle u, v \rangle}{\langle v, v \rangle} v$$

where $\langle \cdot, \cdot \rangle$ is the inner product in V . The vector $u - \operatorname{proj}_v(u)$ is orthogonal to v , i.e.,

$$\langle u - \operatorname{proj}_v(u), v \rangle = 0.$$

Complex Inner Product Spaces (Note): The formula is formally the same:

$$\operatorname{proj}_v(u) = \frac{\langle u, v \rangle}{\langle v, v \rangle} v$$

but now $\langle u, v \rangle$ may be complex. The vector $u - \operatorname{proj}_v(u)$ is orthogonal to v with respect to the complex inner product:

$$\langle u - \operatorname{proj}_v(u), v \rangle = 0.$$

Remark: In complex spaces, the scalar $\frac{\langle u, v \rangle}{\langle v, v \rangle}$ can be complex, but the geometric idea of projection remains the same as in real spaces.

In inner product spaces :

Cauchy–Schwarz Inequality:

$$|\langle u, v \rangle| \leq \|u\| \|v\|, \quad u, v \in V$$

Pythagorean Theorem:

$$u \perp v \implies \|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Triangle Inequality

$$\|u + v\| \leq \|u\| + \|v\|, \quad u, v \in V$$

1) Verify that $\langle u, v \rangle = u_1v_1 + 2u_2v_2$ is an inner product on $\mathbb{R}^2$.

Let $u = (u_1, u_2)$, $v = (v_1, v_2) \in \mathbb{R}^2$. Define

$$\langle u, v \rangle = u_1v_1 + 2u_2v_2.$$

Symmetry:

$$\langle u, v \rangle = u_1v_1 + 2u_2v_2 = v_1u_1 + 2v_2u_2 = \langle v, u \rangle$$

Additivity:

Let $w = (w_1, w_2) \in \mathbb{R}^2$

$$\langle u + w, v \rangle = (u_1 + w_1)v_1 + 2(u_2 + w_2)v_2 = (u_1v_1 + 2u_2v_2) + (w_1v_1 + 2w_2v_2) = \langle u, v \rangle + \langle w, v \rangle$$

Homogeneity:

For $\alpha \in \mathbb{R}$,

$$\langle \alpha u, v \rangle = (\alpha u_1)v_1 + 2(\alpha u_2)v_2 = \alpha(u_1v_1 + 2u_2v_2) = \alpha \langle u, v \rangle$$

Positive Definiteness:

$$\langle u, u \rangle = u_1^2 + 2u_2^2 \geq 0$$

and

$$\langle u, u \rangle = 0 \implies u_1^2 + 2u_2^2 = 0 \implies u_1 = 0, u_2 = 0 \implies u = 0 \text{ (and vice-versa)}$$

All four axioms are satisfied, hence

$$\langle u, v \rangle = u_1v_1 + 2u_2v_2$$

is an inner product on $\mathbb{R}^2$.

2) For weights $c_i > 0$ on $\mathbb{R}^n$, prove that

$$\langle u, v \rangle = \sum_{i=1}^n c_i u_i v_i$$

is an inner product.

Let $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$, and let $c_i > 0$ for $i = 1, 2, \dots, n$.

Define

$$\langle u, v \rangle = \sum_{i=1}^n c_i u_i v_i.$$

Symmetry:

$$\langle u, v \rangle = \sum_{i=1}^n c_i u_i v_i = \sum_{i=1}^n c_i v_i u_i = \langle v, u \rangle$$

Additivity:

Let $w = (w_1, \dots, w_n) \in \mathbb{R}^n$.

$$\langle u + w, v \rangle = \sum_{i=1}^n c_i (u_i + w_i) v_i = \sum_{i=1}^n c_i u_i v_i + \sum_{i=1}^n c_i w_i v_i = \langle u, v \rangle + \langle w, v \rangle$$

Homogeneity:

For $\alpha \in \mathbb{R}$,

$$\langle \alpha u, v \rangle = \sum_{i=1}^n c_i (\alpha u_i) v_i = \alpha \sum_{i=1}^n c_i u_i v_i = \alpha \langle u, v \rangle$$

Positive Definiteness:

$$\langle u, u \rangle = \sum_{i=1}^n c_i u_i^2 \geq 0$$

and

$$\langle u, u \rangle = 0 \implies \sum_{i=1}^n c_i u_i^2 = 0 \implies u_i = 0 \text{ for all } i \implies u = 0 \text{ (and vice-versa)}$$

All four axioms are satisfied, hence

$$\langle u, v \rangle = \sum_{i=1}^n c_i u_i v_i$$

is an inner product on $\mathbb{R}^n$ for any weights $c_i > 0$.

3) Give a counterexample showing that the formula

$$\langle u, v \rangle = \sum_{i=1}^n c_i u_i v_i$$

fails to define an inner product if one weight $c_i = 0$ or $c_i < 0$.

Case 1: One weight is 0: Let $n = 2$, $c_1 = 1$, $c_2 = 0$, and consider $u = (0, 1) \in \mathbb{R}^2$.

$$\langle u, u \rangle = c_1 u_1^2 + c_2 u_2^2 = 1 \cdot 0^2 + 0 \cdot 1^2 = 0$$

but $u \neq 0$.

$\implies$ Positive definiteness fails when a weight is 0.

Case 2: One weight is negative: Let $c_1 = 1$, $c_2 = -1$, and consider $u = (0, 1) \in \mathbb{R}^2$.

$$\langle u, u \rangle = c_1 u_1^2 + c_2 u_2^2 = 1 \cdot 0^2 + (-1) \cdot 1^2 = -1 < 0$$

$\implies$ Positive definiteness fails when a weight is negative.

The formula

$$\langle u, v \rangle = \sum_{i=1}^n c_i u_i v_i$$

defines an inner product only if all weights $c_i > 0$. If any weight is zero or negative, the positive definiteness axiom is violated.

4) On $C[0, 1]$, compute $\|x\|$, $\|x^2\|$, $\langle x, x^2 \rangle$, and the angle between them, where the inner product is defined as

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt.$$

Compute $\|x\|$:

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\int_0^1 x^2 dx} = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$$

Compute $\|x^2\|$:

$$\|x^2\| = \sqrt{\langle x^2, x^2 \rangle} = \sqrt{\int_0^1 x^4 dx} = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}}$$

Compute $\langle x, x^2 \rangle$:

$$\langle x, x^2 \rangle = \int_0^1 x \cdot x^2 dx = \int_0^1 x^3 dx = \frac{1}{4}$$

Compute the angle between x and x^2 :

$$\cos \theta = \frac{\langle x, x^2 \rangle}{\|x\| \|x^2\|} = \frac{1/4}{(1/\sqrt{3})(1/\sqrt{5})} = \frac{\sqrt{15}}{4}$$

$$\theta = \cos^{-1} \left(\frac{\sqrt{15}}{4} \right)$$

5) On $M_{2,2}$, compute $\|A\|$ and check whether the two given matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

are orthogonal with respect to the inner product defined for any $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $B =$

$$\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \text{ by}$$

$$\langle A, B \rangle = a_{11}b_{11} + a_{21}b_{21} + a_{12}b_{12} + a_{22}b_{22}.$$

Norm of A

$$\|A\| = \sqrt{\langle A, A \rangle} = \sqrt{1 \cdot 1 + 3 \cdot 3 + 2 \cdot 2 + 4 \cdot 4} = \sqrt{1 + 9 + 4 + 16} = \sqrt{30}$$

Check orthogonality with B

$$\langle A, B \rangle = 1 \cdot 0 + 3 \cdot (-1) + 2 \cdot 1 + 4 \cdot 0 = 0 - 3 + 2 + 0 = -1$$

Since $\langle A, B \rangle \neq 0$, the matrices A and B are **not orthogonal**.

Let the columns of B be:

$$\begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix}$$

Check orthogonality:

$$\langle A, B \rangle = \mathbf{u}_1 \cdot \mathbf{v}_1 + \mathbf{u}_2 \cdot \mathbf{v}_2 = (1 \cdot 3 + 3 \cdot (-1)) + (2 \cdot 2 + 4 \cdot (-1)) = (3 - 3) + (4 - 4) = 0$$

$$B = \begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix}$$

is orthogonal to A .